Oliver Pierce

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Education

Liberty University, Lynchburg VA

- Double major: BS Computer Science and BS Physics, both expected in May 2031.
- Relevant Coursework Expected by May 2027: Intro to Computer Science, Advanced C++, Technical Communication, Intro to Discrete Mathematics, Calculus 3, Physics 2.

Relevant College Courses During Highschool

- Wake Tech Community College: Calculus 1 & 2, Physics 1 (calculus based). GPA: 4.0
- Judson College: English Composition 1 & 2, Communications. GPA: 4.0

Featured Project: Hexatower

Hexatower is a turn-based multiplayer videogame I developed using Rust and the Bevy Game Engine. More information and a demo can be found at oliverpierce.com. I created this game as a solo developer, without AI.

- Implemented zero trust peer-to-peer multiplayer features using a listen-server architecture.
- Designed a custom hexagonal coordinate system to optimize memory usage and enable multiplayer.
- Leveraged Rust's Type System to make invalid data unrepresentable and prevent misuse of interfaces. This makes code more maintainable and makes it harder to write bugs, saving developer time overall.
- Used the RON format to write custom asset files and load them at runtime through Bevy's assets system.
- Optimized memory layout using data-oriented design and Bevy's entity component system (ECS) model.
- Investigated and fixed bugs in multithreaded and non-deterministic environments.
- Efficiently created 2d and 3d assets in Blender, both for prototyping and release.

Skills

Current Technical Skills: Rust Programming Language, GitHub, Bevy Game Engine, listen-server multiplayer game programming, experience using ECS frameworks, Blender.

Technical Skills by May 2027: C++.

Soft Skills: Curious, Driven, Punctual, Communicative, Teachable, Optimistic.

Achievements and Awards

- Honors Scholar at Liberty University (2026-2030).
- Eagle Scout with Scouting America (2024).
- National Eagle Scout Project of the Year Council Semi-finalist (2025).
- As a Senator, sponsored and passed AI legislation at 2025 TeenPact Congress.

Interests

I'm excited by problems involving mass computation and performance-oriented code. Particularly, I'm interested in simulations, digital twins, machine learning, and distributed systems. Visit oliverpierce.com.